

ORIGINAL ARTICLE

A human-centered framework for miss ICU: A multimodal, evidence-integrated artificial intelligence agent for clinical decision-making in intensive care

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ABSTRACT

Background and aim: The modern Intensive Care Unit (ICU) is a data-rich environment where an overwhelming volume of information can cause cognitive overload for clinicians. Artificial Intelligence (AI) is promising for predictive tasks, but adoption is limited by an implementation gap driven by poor data interoperability, model opacity, and inadequate integration into clinical workflows. This paper presents the “Miss ICU” framework, a novel AI agent designed as an intelligent partner for the critical care team. Its goal is to create an integrated, interpretable, and evidence-based system that augments, rather than replaces, clinical intelligence to improve patient outcomes.

Methods: The framework uses a three-tiered architecture. A Data Harmonization Layer applies standards such as Health Level Seven/Fast Healthcare Interoperability Resources (HL7/FHIR) to build a unified patient data model. An Analytics Engine integrates machine learning for patient-state modeling, Natural Language Processing (NLP) for real-time evidence synthesis from medical literature, and Explainable AI (XAI) for transparency. A human-centered Presentation Layer provides an intuitive dashboard to display patient trajectory and risk.

Results: The envisioned system offers clinicians a holistic view of patient status, including a stability curve to depict clinical trajectory. It aims to generate predictions (e.g., sepsis) with human-readable explanations



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of contributing factors and follows a multi-phase validation pathway ending in a pragmatic randomized controlled trial.

Conclusions: Miss ICU offers a blueprint for next-generation clinical decision support, emphasizing interoperability, interpretability, and integration of patient data with medical evidence to reduce cognitive load, enable proactive care, and improve safety and quality in critical care. (www.actabiomedica.it)

Key words: artificial intelligence, critical care, clinical decision support systems, machine learning, explainable AI, digital health

1. Introduction

The modern Intensive Care Unit (ICU) operates at the frontier of medical science and data complexity. It is an environment where the margin for error is vanishingly small and the cognitive burden on clinicians is immense. Each patient in the ICU generates thousands of data points per hour from a disparate array of sources, including high-frequency physiological waveforms, laboratory results, ventilator parameters, and extensive notations within the Electronic Health Record (EHR) (1). This data deluge has created a well-documented “data-rich, insight-poor” paradox, where the sheer volume of information can obscure critical signals, contributing to diagnostic delays, information oversight, and significant clinician burnout (2-4).

Artificial intelligence (AI) and machine learning (ML) have been heralded as powerful tools to navigate this complexity, with numerous algorithms developed to predict outcomes such as mortality, sepsis onset, and acute kidney injury (5-9). Despite promising results in retrospective studies, the “last mile” of clinical implementation remains largely uncrossed. This implementation gap stems from several persistent challenges: a lack of seamless data interoperability; addressing the digital divide and disparities in access;

improving digital literacy among users, and keeping pace with evolving regulatory requirements; the “black box” nature of many complex models, which engenders clinical mistrust; and a failure to integrate AI-driven insights into the fluid, high-pressure workflows of the ICU (10-13).

Crucially, existing systems rarely bridge the gap between a patient’s immediate data and the global repository of medical knowledge (14). A clinician observing an anomalous trend must still disengage from the patient’s bedside to manually search medical databases—a process too slow for many critical care scenarios. To overcome these barriers, we propose “Miss ICU,” a conceptual framework for a next-generation AI agent designed not as a replacement for clinical expertise, but as a cognitive partner to augment and amplify it (15). In the pursuit of enhancing critical care decision-making, the development of Miss ICU promises to advance patient-specific interventions by integrating evidence-based medicine and real-time data insights, distinguishing itself from prior systems, such as multi-agent sepsis management frameworks (16), through broader contextual understanding, adaptability across diverse ICU scenarios, and comprehensive real-time data processing capabilities.

Statement of significance

Problem or Issue	What is Already Known	What this Paper Adds	Who would benefit from the new knowledge in this paper
ICU clinicians face cognitive overload from overwhelming, fragmented data and lack usable, integrated decision support tools; AI adoption is hindered by issues of interoperability, model opacity, and poor workflow fit.	AI and machine learning have shown promise in retrospective prediction of ICU outcomes, but their practical use is limited by data silos, “black box” algorithms, and failure to synthesize evidence at the bedside.	This paper proposes the “Miss ICU” framework: a human-centered, transparent AI agent that unifies multimodal data, delivers explainable analytics, synthesizes current medical evidence in real time, and provides holistic support via an intuitive interface, with a rigorous validation roadmap.	ICU clinicians, multidisciplinary care teams, patients and families, hospital administrators, and researchers interested in safe, effective integration of AI into critical care environments.

2. Methods: The conceptual framework

The Miss ICU framework was developed through a synthesis of established needs identified in the critical care and medical informatics literature. The design is predicated on three core principles derived from human-computer interaction and implementation science: Interoperability, Interpretability, and Integration into clinical workflow.

2.1. Features

At the heart of the Miss ICU framework is a suite of deeply integrated features designed to create a holistic, real-time digital persona of each patient. This agent uniquely integrates both Generative AI (GAI) and Predictive AI (PAI) within a unified, IoT-connected hospital environment—drawing on real-time data from EHRs, laboratory and imaging systems, and bedside monitors—enabling not only accurate predictions of patient risk and trajectory, but also dynamic generation and sharing of clinically-relevant summaries and recommendations; this comprehensive approach follows the model of advanced cloud-based frameworks that harness both predictive and generative intelligence to enhance operational efficiency and quality of care across healthcare institutions (17). By interfacing directly with hospital information systems (EHR, LIS, PACS), the agent will automatically collate foundational patient data—including personal details, weight, height, admission pathology, and comorbidities—while simultaneously ingesting a live

feed of laboratory results, radiological reports, and other instrumental tests. This static and semi-static data is then fused with high-frequency vital signs, read in real-time from bedside monitors. This fusion of static and dynamic data is the cornerstone of the ‘intelligent ICU’ concept, where the seamless integration of IoT and AI analytics transforms high-volume, real-time data streams into predictive clinical intelligence, enabling a paradigm shift from reactive to proactive patient management (18).

The system’s core analytical engine synthesizes this multimodal information to generate a dynamic ‘pathology curve,’ which visualizes the patient’s clinical trajectory. This visualization is the output of a sophisticated probabilistic model that places the patient on a clinical Gaussian curve of severity, learned from vast historical patient data. As a patient’s real-time data indicates a drift from the population mean toward the tail-end of this distribution—the critical zone—the system’s intuitive traffic-light signal is triggered, shifting from green to yellow, and ultimately to red. Concurrently, the agent generates a second, parallel Gaussian curve representing adherence to the latest evidence-based guidelines, quantifying any deviation between the patient’s current treatment and the recommended standard of care. To make this possible, Miss ICU will have access to therapy, complete EHRs, and “smart” infusion pumps. These dual visual outputs serve critical non-clinical objectives: the Generative AI module translates these complex curves into simple, standardized language for clear communication with relatives (19), while for clinical staff, they

Table 1. Miss ICU Variables Comprehensive Reference Table

Variable Category	Specific Parameters	IoT/EHR Availability	AI Tool Used	Clinical Purpose	Data Frequency
Vital Signs	Heart Rate, Heart Rate Variability	Yes	Both	Cardiac monitoring, arrhythmia detection, hemodynamic assessment	Continuous (1-5 sec)
Vital Signs	Blood Pressure (Invasive/Non-invasive)	Yes	PAI	Hemodynamic stability, shock detection, medication titration	Continuous/Q15min
Vital Signs	Respiratory Rate, Tidal Volume, PEEP	Yes	Both	Ventilator management, weaning protocols, respiratory failure	Continuous
Vital Signs	Oxygen Saturation (SpO2), FiO2	Yes	PAI	Oxygenation assessment, ventilator settings optimization	Continuous
Vital Signs	Core Temperature, Skin Temperature	Yes	PAI	Infection screening, thermoregulation, sepsis detection	Continuous/Q1hr
Vital Signs	Central Venous Pressure (CVP)	Partial	PAI	Volume status assessment, cardiac function evaluation	Continuous (when available)
Laboratory Results	Complete Blood Count (CBC), WBC Differential	Yes	PAI	Infection detection, hematologic assessment, chemotherapy monitoring	Q6-12hrs
Laboratory Results	Basic Metabolic Panel (BMP), Comprehensive Metabolic Panel	Yes	PAI	Electrolyte balance, renal function, acid-base status	Q6-12hrs
Laboratory Results	Arterial Blood Gas (ABG), Venous Blood Gas	Yes	Both	Acid-base balance, oxygenation, ventilation assessment	Q4-8hrs or PRN
Laboratory Results	Lactate, Lactate Clearance	Yes	PAI	Tissue perfusion, sepsis monitoring, shock assessment	Q6-12hrs
Laboratory Results	Coagulation Studies (PT/INR, PTT)	Yes	PAI	Bleeding risk assessment, anticoagulation monitoring	Q12-24hrs
Laboratory Results	Cardiac Enzymes (Troponin, CK-MB, BNP/NT-proBNP)	Yes	PAI	Myocardial infarction diagnosis, heart failure assessment	Q6-8hrs x3, then daily
Laboratory Results	Inflammatory Markers (CRP, Procalcitonin, ESR)	Yes	PAI	Infection detection, sepsis monitoring, treatment response	Daily
Imaging	Chest X-Ray	Partial	Both	Pulmonary pathology detection, device positioning verification	Daily or PRN

Variable Category	Specific Parameters	IoT/EHR Availability	AI Tool Used	Clinical Purpose	Data Frequency
Imaging	CT Scan (Head, Chest, Abdomen)	Partial	Both	Trauma assessment, infection source identification, complications	PRN
Imaging	Echocardiogram (TTE, TEE)	Partial	Both	Cardiac function assessment, fluid status evaluation	PRN
Imaging	Ultrasound (FAST, Lung, Vascular)	No	Both	Point-of-care assessment, fluid evaluation, vascular access	PRN
Microbiology	Blood Cultures, Culture Sensitivities	Yes	GAI	Infection identification, antibiotic susceptibility, sepsis management	On admission, then PRN
Microbiology	Respiratory Cultures (Sputum, BAL, Mini-BAL)	Yes	GAI	Pneumonia diagnosis, VAP prevention, antibiotic selection	On intubation, then PRN
Microbiology	Urine Cultures	Yes	GAI	UTI diagnosis, catheter-associated infections	On admission, then PRN
Microbiology	Molecular Diagnostics (PCR, Rapid Tests)	Partial	GAI	Rapid pathogen identification, viral detection, resistance genes	PRN
Medications	Vasoactive Drugs (Norepinephrine, Dopamine, Vasopressin)	Yes	Both	Hemodynamic support, shock management, blood pressure control	Continuous infusion
Medications	Sedatives (Propofol, Midazolam, Dexmedetomidine)	Yes	PAI	Sedation management, delirium prevention, ventilator synchrony	Continuous infusion
Medications	Antibiotics (β -lactams, Vancomycin, Fluoroquinolones)	Yes	Both	Infection treatment, antibiotic stewardship, resistance prevention	Q6-24hrs
Medications	Insulin Infusion, Blood Glucose Control	Yes	PAI	Glycemic control, hyperglycemia prevention, metabolic management	Continuous infusion
Clinical Guidelines	Sepsis Bundles (SSC Guidelines)	Partial	GAI	Sepsis management standardization, bundle compliance	On sepsis identification
Clinical Guidelines	Ventilator Liberation Protocols	Partial	Both	Weaning protocols, extubation readiness, ventilator-free days	Daily assessment
Clinical Guidelines	DVT Prophylaxis Protocols	Partial	GAI	Thromboembolism prevention, risk stratification	Daily assessment
Clinical Guidelines	Pain-Agitation-Delirium (PAD) Guidelines	Partial	Both	Comfort management, delirium prevention, sedation optimization	Q4-8hrs

(Continued)

Variable Category	Specific Parameters	IoT/EHR Availability	AI Tool Used	Clinical Purpose	Data Frequency
Physical Examination	Glasgow Coma Scale (GCS), Neurological Assessment	Yes	PAI	Consciousness level monitoring, neurological deterioration detection	Q2-4hrs
Physical Examination	Pupil Reactivity, Pupil Size	Partial	PAI	Neurological assessment, intracranial pressure monitoring	Q2-4hrs
Physical Examination	Skin Assessment, Pressure Ulcer Staging	No	GAI	Skin integrity maintenance, pressure injury prevention	Q8-12hrs
Nursing Assessments	Richmond Agitation-Sedation Scale (RASS)	Yes	PAI	Sedation level assessment, delirium screening, comfort management	Q4hrs
Nursing Assessments	Confusion Assessment Method-ICU (CAM-ICU)	Yes	PAI	Delirium detection, cognitive assessment, outcome prediction	Q8-12hrs
Nursing Assessments	Pain Scales (CPOT, BPS, NRS)	Yes	PAI	Pain assessment, comfort management, analgesic titration	Q2-4hrs
Nursing Assessments	Fluid Balance, Input/Output Monitoring	Yes	PAI	Volume status assessment, kidney function, fluid resuscitation	Hourly
Procedures	Mechanical Ventilation Parameters	Yes	Both	Respiratory support, ventilator weaning, lung protection	Continuous
Procedures	Renal Replacement Therapy (CRRT/HD)	Yes	Both	Acute kidney injury management, fluid removal, electrolyte balance	Continuous/Intermittent
Procedures	Central Line Management, PICC Lines	Partial	GAI	Vascular access maintenance, infection prevention, complications	Daily assessment
Procedures	Chest Tube Management	Partial	PAI	Pleural drainage, pneumothorax management, air leak detection	Continuous monitoring
Environmental Factors	Room Temperature, Humidity	Partial	PAI	Patient comfort, infection control, equipment function	Continuous
Environmental Factors	Noise Levels, Light Exposure	No	PAI	Sleep promotion, delirium prevention, circadian rhythm maintenance	Continuous

The table demonstrates how Miss ICU would integrate diverse data sources through both automated IoT systems and manual clinical assessments, using appropriate AI tools for each variable type to support comprehensive critical care decision-making.

provide an objective, at-a-glance summary that dramatically reduces timeframes and mitigates the risk of error during high-pressure handovers. Crucially, Miss ICU contextualizes this patient-specific data by connecting to premier medical databases such as PubMed, Scopus, and Cochrane, providing clinicians with synthesized, evidence-based information on the patient’s condition, potential complications, and recommended treatments. Finally, for a safe and successful discharge, the platform leverages advanced ML algorithms to simultaneously predict both Length of Stay (LOS) and early readmission risk for ICU patients, building upon established methodologies that demonstrate LOS as the most critical predictive feature for readmission outcomes, with potential to achieve superior performance metrics similar to those reported in recent explainable AI frameworks that reached AUROC scores of 0.92 for early readmission prediction (20). Table 1

shows the variables presented by Miss ICU, while Figure 1 shows a concept map of its framework.

2.2. SYSTEM ARCHITECTURE

The proposed architecture is a modular, three-tiered system designed for robustness, scalability, and security.

2.2.1. TIER 1: DATA INGESTION AND HARMONIZATION LAYER

This foundational layer serves as the system’s universal data interface. It is designed to be vendor-agnostic, leveraging modern interoperability standards, primarily HL7 FHIR (Fast Healthcare Interoperability Resources) (21). Notably, the training of Miss ICU leverages recent evidence highlighting that self-supervised pre-training strategies yield the greatest flexibility

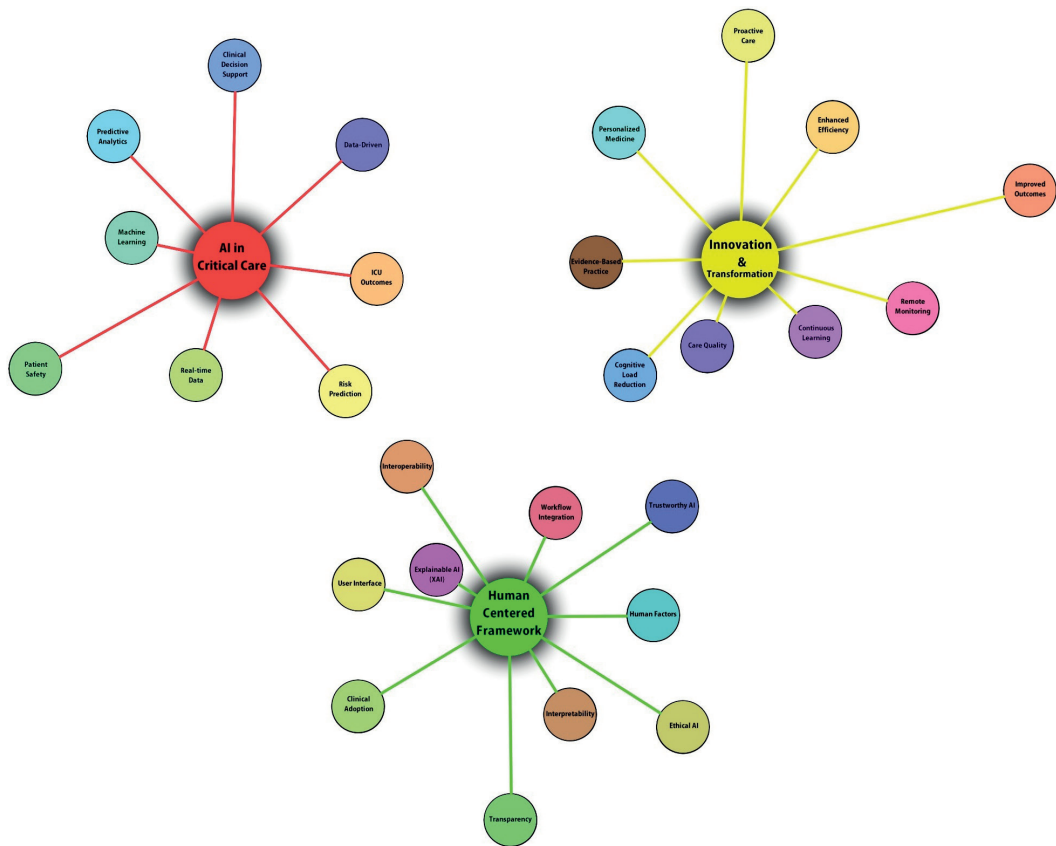


Figure 1. Conceptual map of the Miss ICU Framework. Interconnected domains of AI in critical care, innovation & transformation, and human-centered design principles that underpin the Miss ICU architecture.

and generalizability for AI models tasked with diverse clinical predictions, providing a robust foundation for multitask environments like the ICU, while supervised pre-training may be incorporated for optimizing performance on highly specific outcome predictions, as demonstrated in recent benchmarking studies (22).

- **Data Sources:** Connects via secure APIs to all relevant data sources: EHRs (demographics, comorbidities, notes), Laboratory Information Systems (LIS), Picture Archiving and Communication Systems (PACS), pharmacy records, and real-time data streams from bedside monitors and ventilators.
- **Harmonization Engine:** A key function of this tier is to translate and map incoming data into a coherent, standardized patient data model. It utilizes established medical ontologies (e.g.,

SNOMED-CT for diagnoses, LOINC for lab tests, RxNorm for medications) to resolve semantic heterogeneity, creating a single, machine-readable source of truth for each patient. Figure 2 shows the integrated artificial intelligence system in an intensive care setting.

2.2.2. TIER 2: INTELLIGENT SYNTHESIS AND ANALYTICS ENGINE

This is the cognitive core of Miss ICU, where raw data is transformed into clinical insight.

- **Multimodal Patient State Modeling:** This module employs a hybrid ML approach. Time-series models (e.g., Transformers or LSTMs) are used to analyze high-frequency waveform data, capturing dynamic physiological patterns.

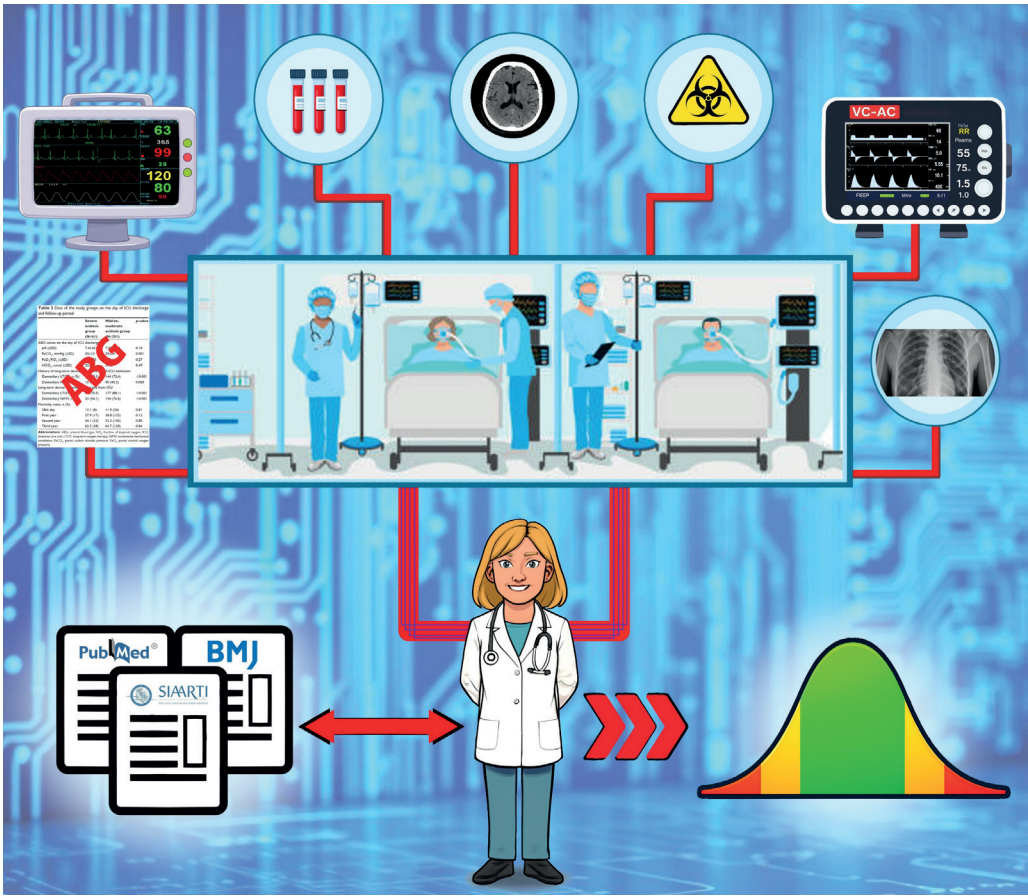


Figure 2. Integrated AI-Driven ICU Data Ecosystem. Schematic depicting Miss ICU’s real-time data fusion: patient monitors, labs, imaging feeds, and evidence sourcing converge to drive predictive analytics and visualization.

These are combined with gradient-boosted models (e.g., XGBoost) that integrate static and low-frequency data (e.g., admission details, daily labs). The output is a dense vector representation of the patient's comprehensive clinical state at any given moment.

- **Dynamic Evidence-Synthesis Module:** This module leverages state-of-the-art Natural Language Processing (NLP). When a patient's profile is updated (e.g., a new diagnosis, a critical lab value), this module automatically formulates a query to trusted medical databases (PubMed, Scopus, Embase, Cochrane). It retrieves relevant abstracts and utilizes transformer-based summarization models to generate concise, evidence-based bullet points on diagnostics, prognostics, and therapeutic options, which are then presented to the clinician.
- **Predictive Analytics and Explainable AI (XAI) Module:** This module uses the patient state vector to drive predictive models for key clinical endpoints. Every prediction is mandatorily processed by an XAI layer (e.g., SHAP - SHapley Additive exPlanations, or LIME - Local Interpretable Model-agnostic Explanations) to generate human-readable justifications, enhancing transparency and trust.

2.2.3. TIER 3: CLINICAL PRESENTATION AND INTERACTION LAYER

This is the human-centered user interface, designed for at-a-glance comprehension and minimal workflow disruption.

- **The “Traffic Light” Stability Dashboard:** The central feature is a visualization of the patient's stability score over time, plotted as a continuous curve and color-coded (Green: stable; Yellow: potential instability, requires attention; Red: high risk of deterioration). This focuses on the patient's trajectory, which is often more informative than a single snapshot. This score is generated by a sophisticated state-space deep learning engine, inspired by state-of-the-art models like APRICOT-M, which dynamically

assesses patient acuity in real-time (23). Unlike traditional scoring systems that offer a static view of risk, Miss ICU's model continuously analyzes streams of clinical data to predict not just mortality but also the imminent need for specific life-sustaining therapies within the next four hours. The color-coded trajectory thereby provides clinicians with an actionable, forward-looking assessment, allowing for earlier and more targeted interventions, a methodology proven to be highly effective and clinically relevant in recent studies. This system is showed in Figure 3.

- **Unified View:** The dashboard integrates all relevant information—predictions, their explanations, and the synthesized evidence—into a single, context-aware interface, eliminating the need for clinicians to “app-switch.”
- **Interactive Querying:** The interface allows for direct interaction, enabling clinicians to pose questions like, “Show evidence for ARDSnet protocol in a patient with these characteristics,” or to run “what-if” simulations on the potential effects of interventions.

2.3. Ethical considerations and governance

The development and deployment of Miss ICU must be governed by a rigorous ethical framework.

- **Bias and Fairness:** The training datasets will be carefully audited for demographic and socioeconomic biases. The models' performance will be evaluated across different patient subgroups to ensure equitable outcomes (24).
- **Accountability:** Miss ICU is a decision-support tool, and the final accountability for clinical decisions remains with the human clinician. The system's logs and the XAI outputs are designed to be fully auditable to understand the information presented to the clinician at the time of a decision.
- **Data Privacy and Security:** The system will be designed for on-premise deployment within a hospital's secure firewall, and all data will be handled in strict compliance with HIPAA and GDPR regulations.

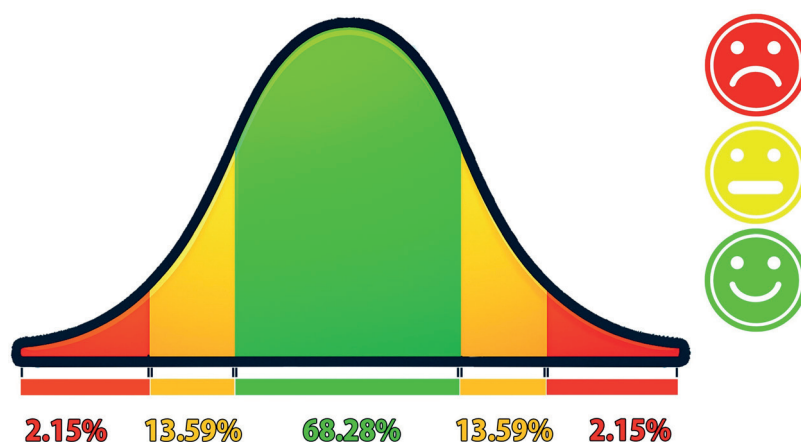


Figure 3. Patient Stability Gaussian Curve with Traffic-Light Risk Zones.

A normal distribution of patient acuity scores segmented into green (68.3%), yellow (13.6% each), and red (2.15% each) zones, mapped to smiley-face risk indicators for at-a-glance stability assessment.

This commitment to a rigorous ethical framework directly aligns with the expert consensus that safe AI integration into critical care is contingent upon a human-centric philosophy, transparent governance, and a model of shared accountability among clinicians, developers, and regulatory bodies to ensure that technology enhances, rather than undermines, patient care (25).

2.3.1 FRAMEWORK FOR TRANSPARENCY: LAYERED MODEL CARDS

To operationalize our commitment to transparency and accountability, the Miss ICU framework adopts the use of layered Model Cards. This methodology provides a structured and multi-faceted approach to documenting the system's complex AI components, ensuring that clear, appropriate information is available to all stakeholders, from clinicians to patients and their families (26, 27). The proposed model card structure consists of three distinct layers:

- **Layer 1: High-Level Summary Label.** Designed for rapid comprehension by clinicians, patients, and families, this layer functions as an AI “nutrition label.” It will use standardized graphical elements, including the framework’s
- **Layer 2: Detailed Technical and Clinical Report.** This layer is targeted at ICU staff, hospital IT, and administrative personnel. It provides comprehensive documentation on the underlying AI architecture, including the hybrid use of time-series models and gradient-boosted trees. It details the specific input variables and output predictions, reports granular quantitative analysis from the validation roadmap (including fairness and bias assessments across patient subgroups), and explains the use of integrated XAI methods (e.g., SHAP, LIME) for model interpretability.
- **Layer 3: Verifiable, Computable Artifact.** To ensure full auditability, reproducibility, and alignment with FAIR (Findable, Accessible, Interoperable, and Reusable) data principles,

native traffic-light visualizations, to communicate the model’s specific intended use, key functions (e.g., instability prediction, data summarization), and primary performance metrics (e.g., AUROC). Crucially, this layer will also feature explicit warnings regarding model limitations and the intended patient population, reinforcing that the system is a decision-support tool, not a replacement for clinical judgment.

the third layer provides deep technical specifications in a machine-readable format (28). This includes an ontology-based representation of the model's components (leveraging standards like MCRO), detailed descriptions of the training datasets (e.g., MIMIC-IV) and preprocessing pipelines, and links to version-controlled repositories to support regulatory validation and further research.

By implementing this layered model card, the Miss ICU framework provides a robust mechanism for safety, trust, and governance, setting a new standard for transparency in critical care AI systems (26, 27).

3. Results: Envisioned clinical utility

The Miss ICU framework is designed to yield tangible improvements in the ICU. We envision its utility in three main areas:

1. **Reduced Cognitive Load:** By automating data aggregation, synthesis, and prioritization, Miss ICU allows clinicians to focus their cognitive energy on complex decision-making rather than on information retrieval (29). By harnessing the concept of 'intelligent time,' Miss ICU streamlines data-intensive tasks and enables clinicians to refocus their attention on patient-centered care, enhancing proactive decision-making, shared interactions, and holistic treatment, as envisioned in frameworks promoting AI-driven healthcare innovations (30).
2. **Proactive and Preemptive Care:** The trajectory-based stability monitoring and early warnings are designed to shift care from a reactive to a proactive paradigm, allowing for interventions before a patient overtly decompensates (31).
3. **Enhanced Evidence-Based Practice:** By bringing tailored, evidence-based knowledge directly to the bedside, the system facilitates the consistent application of best practices and supports personalized medicine. Figure 4 shows possible examples of use in clinical practice.

4. Discussion

The novelty of the Miss ICU framework lies not in a single algorithmic breakthrough but in its holistic, integrated design that directly confronts the primary barriers to AI adoption in critical care. Unlike tools that offer a single prediction, Miss ICU proposes a comprehensive cognitive partnership with the clinician. This aligns with the growing consensus that the responsible and phased implementation of artificial intelligence in the ICU requires a strategic framework that proactively addresses technical, organizational, and clinical barriers to safely improve patient outcomes and clinician workflows (32).

The emphasis on explainability is critical. In a high-stakes environment, a "black box" prediction is clinically inert. By providing a "why" alongside the "what," the XAI module fosters trust, allows for clinical validation of the AI's reasoning, and provides a valuable educational function for trainees (33). Furthermore, the dynamic integration of medical literature represents a significant leap beyond current decision support systems, which typically rely on static, manually curated knowledge bases (34).

Limitations: As a conceptual framework, Miss ICU has not yet been implemented or validated. The technical challenges of achieving seamless, real-time interoperability across all hospital systems are substantial. Furthermore, the social and organizational challenges of integrating such a tool into established clinical workflows and cultures will require careful change management and extensive user training.

Validation roadmap

A staged, "risk-gated" validation strategy is essential to move Miss ICU from concept to routine bedside use while satisfying scientific, ethical, and regulatory expectations (FDA SaMD, EU MDR, IEC 62304). Each phase has explicit entry/exit criteria, success metrics, and predefined "stop/go" thresholds to prevent premature deployment (Table 2).

ITERATION & GOVERNANCE

Phases are iterative; model recalibration or UI redesign feeds back to earlier stages when thresholds

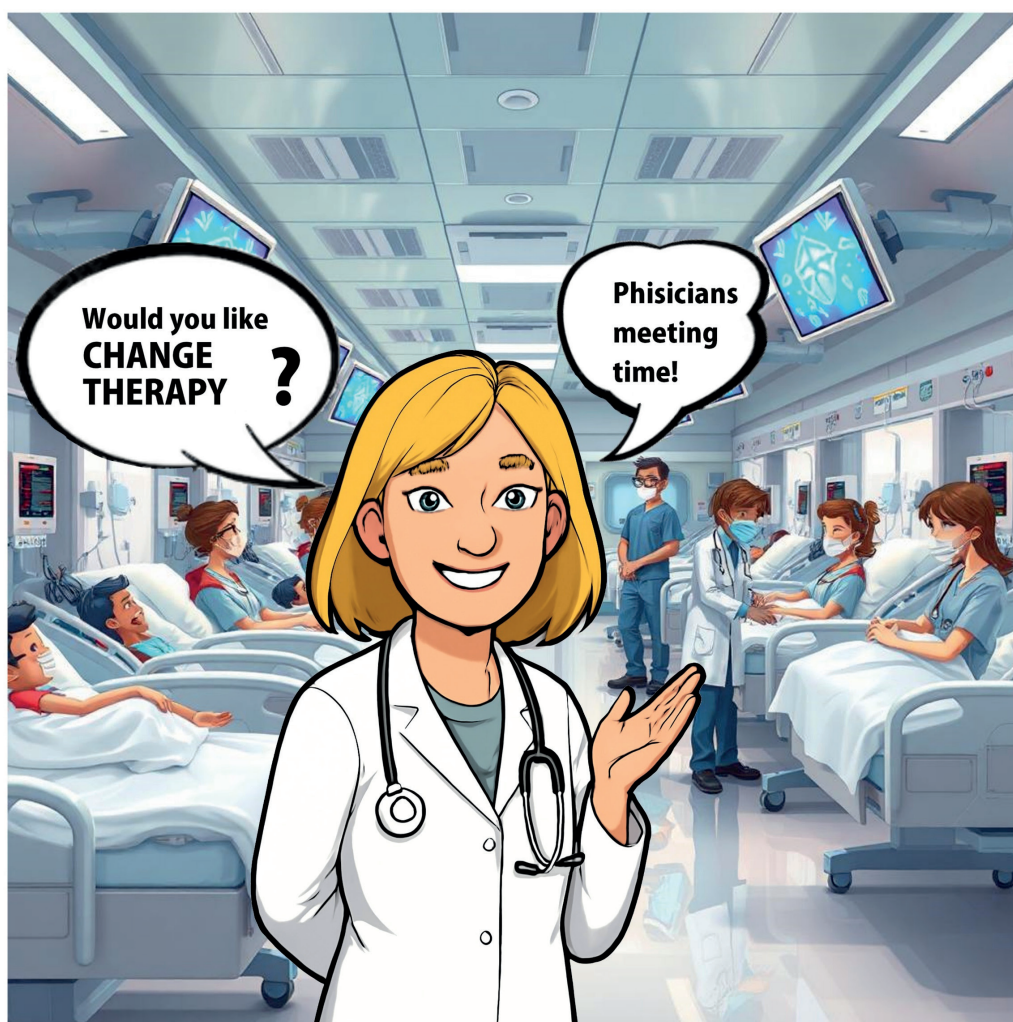


Figure 4. Illustrative Scenario of AI-Enhanced ICU Workflow. Cartoon depiction of Miss ICU agent prompting therapeutic decisions and coordinating physician rounds in a busy critical care ward

are not met. All code is version-controlled, and model updates are locked during prospective phases to prevent dataset shift. Ethics approval, informed consent (opt-out for minimal-risk observational phases), and data-protection impact assessments are obtained at each site.

REAL LIFE SURVEILLANCE

After successful RCT and regulatory clearance, Miss ICU enters a continuous-learning lifecycle:

automated drift detection, quarterly performance audits, user-reported incident tracking, and annual re-certification, ensuring sustained safety and efficacy across evolving patient populations and clinical practices.

5. Future directions

The vision for Miss ICU extends beyond the walls of the intensive care unit, aiming to support the

Table 2. Validation Roadmap Phases

Phase	Goal	Setting & Scale	Key Activities	Primary Metrics	Success Thresholds	Gatekeeper(s)
1. Retrospective Model Development & Validation	Demonstrate technical feasibility and baseline accuracy.	De-identified public databases (MIMIC-IV, eICU, Amsterdam UMCdb; $\approx 100\,000$ ICU stays).	Clean & harmonize data to Miss ICU ontology. Train/validate prediction models via nested cross-validation and temporal hold-out. Calibrate uncertainty estimates. Benchmark vs. published models.	AUROC, AUPRC, Brier score, calibration slope, SHAP global feature importance stability.	AUROC ≥ 0.85 for instability prediction; calibration slope 0.9-1.1.	Data-science lead + independent statistician.
2. In-Silico (“Digital-Twin”) Simulation	Stress-test full stack—including UI—without patient risk.	High-fidelity simulation lab; synthetic waveform generator; 30 virtual “patient days.”	Replay historical data streams in real time. Clinicians interact with Miss ICU while blinded to true outcomes. Cognitive walkthrough & think-aloud usability testing.	NASA-TLX workload, SUS usability score, alert precision/recall, end-to-end latency.	SUS ≥ 80 ; alert precision $\geq 70\%$; latency < 5 s.	Human-factors team + ICU nurse champion.
3. Prospective “Silent-Mode” Observational Study	Validate performance on live data; collect ecological usability feedback.	One tertiary ICU, 3 months, ≈ 300 admissions.	Miss ICU runs in parallel; outputs hidden from care team. Compare predictions to observed events. Log integration failures & data-flow gaps.	Real-time AUROC, calibration drift, missing-data rate, system uptime, clinician Net Promoter Score (NPS) survey.	No critical downtime > 30 min; AUROC drop $< 5\%$ vs. Phase 1.	Local IRB + hospital IT security.
4. Pilot Open-Label Feasibility Trial	Assess safety, workflow impact, and surrogate outcomes with user access.	Same ICU, 4-6 weeks, ≈ 120 patients.	Clinicians view Miss ICU dashboard; all actions logged. Collect time-to-recognition of instability, intervention timeliness. • Monitor for automation bias/adverse events.	Safety event rate, task-completion time, Page-B test for handover accuracy, REBA (Rapid Entire Body Assessment) for ergonomic impact.	No increase in adverse events; $\geq 15\%$ reduction in time-to-intervention vs. historic controls.	Data Safety Monitoring Board.
5. Pragmatic Multicentre Randomized Controlled Trial and Full-Scale implementation	Establish clinical efficacy and generalizability.	≥ 5 ICUs (academic + community), cluster RCT, 2 $\times$ 6-month crossover, $\approx 3\,000$ patients.	Units randomized to Miss ICU-enabled care vs. usual care. Stratify by case mix index. Mixed-methods process evaluation (CFIR framework).	Primary: ICU mortality. Secondary: hospital LOS, ventilator-free days, sepsis bundle compliance, clinician burnout (MBI), handover error rate, family-communication satisfaction.	Relative risk reduction in ICU mortality $\geq 10\%$ or absolute LOS reduction ≥ 1 day with $P < 0.05$; no signal of harm.	Independent DSMB, regulatory submission package (FDA 510(k)/EU CE-Mark).

patient's full recovery journey by transforming into a personalized post-discharge monitoring agent. Upon discharge, the framework can leverage its deep, longitudinal understanding of the patient's clinical history to generate a dynamic and personalized care plan. This plan would be accessible to patients and their caregivers via a secure, patient-facing mobile application.

A key innovation in this phase would be the integration with consumer-grade wearable devices (e.g., smartwatches, connected scales, blood pressure cuffs) and digital platforms for Patient-Reported Outcome (PRO) surveys. This would enable the continuous, real-time collection of physiological data such as heart rate variability, sleep patterns, and activity levels, alongside subjective feedback on their recovery. The system would employ specialized anomaly detection models to monitor these data streams for subtle signs of clinical deterioration or to identify key recovery milestones (35).

Crucially, this remote monitoring framework is designed to bridge the critical and often fragmented gap between intensive care and primary care. By integrating with outpatient EHRs, it can provide automated alerts and synthesized reports directly to community health providers or trigger a telehealth consultation if adverse trends are detected. This ensures that high-risk patients receive proactive and timely interventions, aiming to reduce hospital readmissions, enhance post-discharge Quality of Life (QoL), and establish a true continuum of care. This approach requires an unwavering commitment to data security, ensuring all patient data is encrypted and handled in compliance with privacy regulations like HIPAA, with patient consent as the cornerstone of the system.

6. Conclusions

The Miss ICU framework represents a significant step forward, offering a comprehensive and robust blueprint for the next generation of clinical AI in critical care. Its primary contribution is the unification of three critical pillars: seamless data integration from disparate sources, advanced and interpretable predictive modeling, and a rigorous, multi-phase validation

roadmap. By systematically addressing the persistent challenges of interoperability, clinical trust, and evidence synthesis, it outlines a clear, responsible path toward creating an AI agent that can serve as a trusted and indispensable partner for the critical care team.

The emphasis on a phased validation strategy—from retrospective analysis to a multi-center randomized controlled trial—is central to this framework. It provides the essential safeguards and evidence-based checkpoints necessary to ensure patient safety, clinical efficacy, and regulatory alignment before deployment.

The ultimate goal is to leverage the power of machine intelligence not to replace, but to augment human expertise, creating a symbiotic partnership between clinician and algorithm. By offloading the cognitive burden of continuous data synthesis and risk stratification, Miss ICU frees clinicians to focus on what they do best: complex decision-making, hands-on procedures, and compassionate patient care. This framework paves the way for a safer, more efficient, and more personalized ICU environment, ultimately helping to deliver the best possible outcome for every patient.

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Conflict of Interest: None.

Authors Contribution: EGB, MR, EC and RL contributed to the conceptualization and design of this study. MR was the major contributor to the writing of the manuscript and drafted the initial content. EC, and VB revised the manuscript. EGB and VB provided critical insights into the clinical applications of artificial intelligence in Intensive Care settings and reviewed the manuscript for clinical accuracy. VB, RL and EC contributed to the ethical and policy analyses, ensuring the manuscript addressed key considerations for responsible adoption. RL create the figures and offered expertise in data-driven methodologies, shaping the technical discussions.

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